

GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX - DTC:
TRANSMISSION CONTROL MODULE [TCM] [G1796245]

DESCRIPTION AND OPERATION

TRANSMISSION CONTROL MODULE [TRANSMISSION CONTROL MODULE
[TCM]]

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- Guided Diagnostics must be followed and the repair advised by the process completed. Failure to do so may result in the rejection of any warranty claim made.
- If a control module or a component may be at fault and the vehicle is under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval program is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the Jaguar Land Rover approved diagnostic equipment).
- When performing voltage or resistance tests, always use a digital multimeter that has the resolution ability to view 3 decimal places. For example, on the 2 volts range can measure 1mV or 2 K Ohm range can measure 1 Ohm. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If Diagnostic Trouble Codes (DTC) are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check Jaguar Land Rover (JLR) claims submission system for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and complete the recommendations as required.
- If a failure condition is reported indicating the need to renew the transmission assembly on a vehicle that remains under manufacturer warranty, an authorisation request must first go through the warranty prior approval program before any work is begun.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be set in the Transmission Control Module (Transmission Control Switch (TCS)). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual. For additional information, refer to: Diagnostics - Vehicles With: 8HP70 8-Speed Automatic Transmission AWD (307-01 Automatic Transmission/Transaxle - Vehicles With: 8HP70 8-Speed Automatic Transmission AWD, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B108E-02	Display - General signal failure	■ Gear display signal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u>. Perform a road test and check that the displayed gear matches the engaged gear. If the fault persists, install a new transmission control module
C1A88-64	High Pressure Switching Valve #1 - Signal plausibility failure	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0219-86	Engine Overspeed Condition - Signal invalid	■ Missing/invalid data from the powertrain control module	■ Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
P0560-21	System Voltage - Signal amplitude < minimum	■ Transmission control module power or ground circuit open circuit, high resistance ■ Battery/charging system fault	■ Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ■ Refer to the relevant section of the workshop manual and check the battery and charging system. Rectify as necessary
P0562-1C	System Voltage Low - Circuit voltage out of range	■ Transmission control module power or ground circuit open circuit, high resistance ■ Battery/charging system fault	■ Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ■ Refer to the relevant section of the workshop manual and check the battery and charging system. Rectify as necessary
P0562-21	System Voltage Low - Signal amplitude < minimum	■ Transmission control module power or ground circuit open circuit, high resistance ■ Battery/charging system fault	■ Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ■ Refer to the relevant section of the workshop manual and check the battery and charging system. Rectify as necessary
P0563-22	System Voltage High - Signal amplitude > maximum	■ Battery/charging system fault	■ Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the relevant section of the workshop manual and check the battery and charging system. Rectify as necessary
P0601-41	Internal Control Module Memory Checksum Error - General checksum failure	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0606-04	Control Module Processor - System internal failures	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0606-11	Control Module Processor - Circuit short to ground	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0606-62	Control Module Processor - Signal compare failure	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0606-64	Control Module Processor - Signal plausibility failure	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0613-04	<u>TCS</u> Processor - System internal failures	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0613-06	<u>TCS</u> Processor - Algorithm based failures	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0613-11	<u>TCS</u> Processor - Circuit short to ground	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0613-12	<u>TCS</u> Processor - Circuit short to battery	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0613-13	<u>TCS</u> Processor - Circuit open	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0613-15	<u>TCS</u> Processor - Circuit short to battery or open	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0613-18	<u>TCS</u> Processor - Circuit current below threshold	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P061B-02	Internal Control Module Torque Calculation Performance - General signal failure	■ Transmission control module power or ground circuit open circuit, high resistance ■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P061B-64	Internal Control Module Torque Calculation Performance - Signal plausibility failure	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P062F-04	Internal Control Module EEPROM Error - System internal failures	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0641-64	Sensor Reference Voltage A Circuit/Open - Signal plausibility failure	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0642-21	Sensor Reference Voltage A Circuit Low - Signal amplitude < minimum	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0643-22	Sensor Reference Voltage A Circuit High - Signal amplitude > maximum	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0657-13	Actuator Supply Voltage A Circuit/Open - Circuit open	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0657-64	Actuator Supply Voltage A Circuit/Open - Signal plausibility failure	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0658-11	Actuator Supply Voltage A Circuit Low - Circuit short to ground	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0659-12	Actuator Supply Voltage A Circuit High - Circuit short to battery	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0666-64	Control Module Internal Temperature Sensor A Circuit - Signal plausibility failure	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>
P0667-01	Control Module Internal Temperature Sensor A Range/Performance - General electrical failure	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>
P0668-21	Control Module Internal Temperature Sensor A Circuit Low - Signal amplitude < minimum	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0669-22	Control Module Internal Temperature Sensor A Circuit High - Signal amplitude > maximum	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>
P06AC-01	Control Module Internal Temperature Sensor B Range/Performance - General electrical failure	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>
P06AC-64	Control Module Internal Temperature Sensor B Range/Performance - Signal plausibility failure	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>
P06AD-22	Control Module Internal Temperature Sensor B Circuit Low - Signal amplitude > maximum	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>
P06AE-21	Control Module Internal Temperature Sensor B Circuit High - Signal amplitude < minimum	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>
P0700-02	Transmission Control System (Malfunction Indicator Lamp (MIL) Request) - General signal failure	Transmission system fault	NOTE: If other transmission system <u>DTCs</u> are set, perform the corrective actions for those <u>DTCs</u> first and retest. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test
P0700-83	Transmission Control System (<u>MIL</u> Request) - Value of signal protection calculation incorrect	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0702-98	Transmission Control System Electrical - Component or system over temperature	- Transmission cooling system fault - Transmission control module internal failure	- Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Check the automatic transmission cooling system. Rectify as necessary - If the fault persists, install a new transmission control module. Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Learn Vehicle Identification Number (VIN) (0x0404) -

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0710-04	Transmission Fluid Temperature Sensor A Circuit - System internal failures	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>
P0710-13	Transmission Fluid Temperature Sensor A Circuit - Circuit open	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>
P0710-27	Transmission Fluid Temperature Sensor A Circuit - Signal rate of change above threshold	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>
P0710-64	Transmission Fluid Temperature Sensor A Circuit - Signal plausibility failure	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>
P0712-11	Transmission Fluid Temperature Sensor A Circuit Low - Circuit short to ground	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>
P0713-12	Transmission Fluid Temperature Sensor A Circuit High - Circuit short to battery	See note	NOTE: This <u>DTC</u> is for event information only and does not indicate a fault. Ignore this <u>DTC</u>
P0715-12	Input Shaft Speed Sensor A Circuit - Circuit short to battery	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0715-64	Input Shaft Speed Sensor A Circuit - Signal plausibility failure	Input sensor signal implausible	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. Check datalogger signals - Input Speed (0x1E72) - Transmission Output Shaft Speed (0x1E68) - for plausibility. If the fault persists, install a new transmission control module
P0716-14	Input Shaft Speed Sensor A Circuit Range/Performance - Circuit short to ground or open	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0716-27	Input Shaft Speed Sensor A Circuit Range/Performance - Signal rate of change above threshold	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0720-12	Output Shaft Speed Sensor Circuit - Circuit short to battery	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0720-14	Output Shaft Speed Sensor Circuit - Circuit short to ground or open	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0721-02	Output Shaft Speed Sensor Circuit Range/Performance - General signal failure	Output shaft speed sensor implausible	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. Check datalogger signals - Input Speed (0x1E72) - Transmission Output Shaft Speed (0x1E68) - for plausibility. If the fault persists, install a new transmission control module
P0721-27	Output Shaft Speed Sensor Circuit Range/Performance - Signal rate of change above threshold	Output shaft speed sensor signal implausible	NOTE: This <u>DTC</u> may be induced by an anti-lock braking system fault that allows the brakes to lock during braking. Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index. Clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0721-62	Output Shaft Speed Sensor Circuit Range/Performance - Signal compare failure	Output shaft speed sensor circuit signal compare failure	NOTE: This <u>DTC</u> does not apply to vehicles from 2015 model year. Check the transmission control module connector and the power/ground circuits to the transmission control module. Using the Jaguar Land Rover approved diagnostic equipment, complete a Controller Area Network (CAN) network integrity test. Clear the <u>DTCs</u> and retest. If the problem persists, renew the transmission control module as required
P0721-64	Output Shaft Speed Sensor Circuit Range/Performance - Signal plausibility failure	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0725-81	Engine Speed Input Circuit - Invalid serial data received	Missing/invalid data from the powertrain control module	Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
P0729-07	Gear 6 Incorrect Ratio - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0730-00	Incorrect Gear Ratio - No sub type information	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P0731-07	Gear 1 Incorrect Ratio - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P0732-07	Gear 2 Incorrect Ratio - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P0733-07	Gear 3 Incorrect Ratio - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P0734-07	Gear 4 Incorrect Ratio - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P0735-07	Gear 5 Incorrect Ratio - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P0736-07	Reverse Incorrect Ratio - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P0736-64	Reverse Incorrect Ratio - Signal plausibility failure	Automatic transmission internal failure	NOTE: This <u>DTC</u> does not apply to vehicles from 2015 model year. Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0741-07	Torque Converter Clutch Circuit Performance / Stuck Off - Mechanical failures	■ Torque converter or automatic transmission failure	■ Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new torque converter. Clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P0745-11	Pressure Control Solenoid "A" - Circuit short to ground	■ Transmission control module power supply fault ■ Transmission ground connection fault ■ Hydraulic Impulse Storage (HIS) oil accumulator harness fault ■ Automatic transmission fluid level not as specified or in poor condition	■ Prioritised steps to perform: ■ Check the 12v battery charge and condition. Record values. ■ Using the Jaguar Land Rover approved diagnostic equipment, read all DTCs and investigate any DTCs relating to battery voltage supply. Rectify as required ■ Check the security and integrity of the transmission ground point and check resistance from transmission ground to vehicle ground. Rectify as required ■ Check the transmission VBATT and IGN circuits to confirm an appropriate voltage supply to transmission connector. Rectify as required ■ Check the transmission connector(s) for damage and/or water ingress and rectify as required ■ Refer to the workshop manual, remove the transmission fluid pan and check the security of the Hydraulic Impulse Storage (HIS) oil accumulator electrical connector. Check the harness of the Hydraulic Impulse Storage (HIS) oil accumulator is correctly routed and not pinched or damaged. Rectify as required ■ Check the transmission fluid level using the Jaguar Land Rover approved diagnostic equipment to run the 8-speed ZF - Automatic Transmission Fluid Level Check Application. Refer to the relevant section of the workshop manual and check the automatic transmission fluid quality. For additional information, refer to the Transmission Oil Color Chart in the relevant section of the workshop manual. If the condition is unsatisfactory, raise a Technical Assist (TA) for further support ■ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest. If the fault persists, raise a Technical Assist (TA) for further support
P0745-12	Pressure Control Solenoid A - Circuit short to battery	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0745-13	Pressure Control Solenoid A - Circuit open	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0745-14	Pressure Control Solenoid A - Circuit short to ground or open	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0745-64	Pressure Control Solenoid A - Signal plausibility failure	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0745-93	Pressure Control Solenoid A - No operation	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0745-94	Pressure Control Solenoid A - Unexpected operation	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0748-11	Pressure Control Solenoid A Electrical - Circuit short to ground	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0748-12	Pressure Control Solenoid A Electrical - Circuit short to battery	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0748-13	Pressure Control Solenoid A Electrical - Circuit open	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P076F-07	Gear 7 Incorrect Ratio - Mechanical failures	■ Automatic transmission internal failure	■ Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P077E-02	Transmission Fluid Temperature Measurement System - Multiple Sensor Correlation - General signal failure	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0780-62	Shift Error - Signal compare failure	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0780-93	Shift Error - No operation	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0780-94	Shift Error - Unexpected operation	■ Automatic transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0795-04	Pressure Control Solenoid C - System internal failures	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P07AD-07	Transmission Friction Element F Stuck On - Mechanical failures	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P07D9-07	Gear 8 Incorrect Ratio - Mechanical failures	■ Automatic transmission internal failure	■ Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P07DC-07	Incorrect Shift from Gear 1 - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P07DD-07	Incorrect Shift from Gear 2 - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P07DE-07	Incorrect Shift from Gear 3 - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P07DF-07	Incorrect Shift from Gear 4 - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P07E0-07	Incorrect Shift from Gear 5 - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P07E1-07	Incorrect Shift from Gear 6 - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P07E2-07	Incorrect Shift from Gear 7 - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P07E3-07	Incorrect Shift from Gear 8 - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P0801-94	Reverse Inhibit Control Circuit/Open - Unexpected operation	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0810-04	Clutch Position Control Error - System internal failures	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u>. Perform a road test and check that the displayed gear matches the engaged gear. If the fault persists, install a new transmission control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P084F-11	Park/Neutral Switch Output Circuit - Circuit short to ground	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P084F-12	Park/Neutral Switch Output Circuit - Circuit short to battery	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P084F-13	Park/Neutral Switch Output Circuit - Circuit open	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P084F-14	Park/Neutral Switch Output Circuit - Circuit short to ground or open	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P084F-29	Park/Neutral Switch Output Circuit - Signal invalid	NOTE: Circuit reference - PARK - ■ Park/neutral output signal circuit short circuit to ground, short circuit to power ■ Transmission control module internal failure	■ Refer to the electrical circuit diagrams and check the park/neutral output signal circuit for short circuit to ground, short circuit to power. Repair the wiring harness as necessary ■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0850-02	Park/Neutral Switch Input Circuit - General signal failure	■ Emergency park release deployed ■ Parklock mechanism failure ■ Transmission control module internal failure	■ Check the integrity of the emergency park release. Rectify as necessary ■ Check the integrity of the parklock mechanism. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. Check the parklock mechanism by engaging and disengaging the parklock several times. Rectify as necessary ■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0850-29	Park/Neutral Switch Input Circuit - Signal invalid	■ Emergency park release deployed ■ Parklock mechanism failure ■ Transmission control module internal failure	■ Check the integrity of the emergency park release. Rectify as necessary ■ Check the integrity of the parklock mechanism. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. Check the parklock mechanism by engaging and disengaging the parklock several times. Rectify as necessary ■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0851-01	Park/Neutral Switch Input Circuit Low - General electrical failure	- Emergency park release deployed - Parklock mechanism failure - Transmission control module internal failure	- Check the integrity of the emergency park release. Rectify as necessary - Check the integrity of the parklock mechanism. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. Check the parklock mechanism by engaging and disengaging the parklock several times. Rectify as necessary - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0852-01	Park/Neutral Switch Input Circuit High - General electrical failure	- Emergency park release deployed - Parklock mechanism failure - Transmission control module internal failure	- Check the integrity of the emergency park release. Rectify as necessary - Check the integrity of the parklock mechanism. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. Check the parklock mechanism by engaging and disengaging the parklock several times. Rectify as necessary - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0942-07	Hydraulic Pressure Unit - Mechanical failures	Automatic transmission internal failure	NOTE: This <u>DTC</u> applies to vehicles equipped with the auto stop/start system. Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P0942-62	Hydraulic Pressure Unit - Signal compare failure	Mismatch between the engine and transmission auto stop/start system signals	NOTE: This <u>DTC</u> applies to vehicles equipped with the auto stop/start system. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. Check the engine and transmission auto stop/start system signals when performing an auto stop/start. If the fault persists, install a new transmission control module
P0965-64	Pressure Control Solenoid B Control Circuit Range/Performance - Signal plausibility failure	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0965-93	Pressure Control Solenoid B Control Circuit Range/Performance - No operation	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0965-94	Pressure Control Solenoid B Control Circuit Range/Performance - Unexpected operation	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0966-11	Pressure Control Solenoid B Control Circuit Low - Circuit short to ground	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0966-12	Pressure Control Solenoid B Control Circuit Low - Circuit short to battery	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0966-14	Pressure Control Solenoid B Control Circuit Low - Circuit short to ground or open	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0972-13	Shift Solenoid A Control Circuit Range/Performance - Circuit open	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0972-64	Shift Solenoid A Control Circuit Range/Performance - Signal plausibility failure	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0972-93	Shift Solenoid A Control Circuit Range/Performance - No operation	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0972-94	Shift Solenoid A Control Circuit Range/Performance - Unexpected operation	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0973-11	Shift Solenoid A Control Circuit Low - Circuit short to ground	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0973-18	Shift Solenoid A Control Circuit Low - Circuit current below threshold	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0974-12	Shift Solenoid A Control Circuit High - Circuit short to battery	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0974-19	Shift Solenoid A Control Circuit High - Circuit current above threshold	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0975-13	Shift Solenoid B Control Circuit Range/Performance - Circuit open	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0975-64	Shift Solenoid B Control Circuit Range/Performance - Signal plausibility failure	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0975-93	Shift Solenoid B Control Circuit Range/Performance - No operation	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0975-94	Shift Solenoid B Control Circuit Range/Performance - Unexpected operation	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0976-11	Shift Solenoid B Control Circuit Low - Circuit short to ground	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0976-18	Shift Solenoid B Control Circuit Low - Circuit current below threshold	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0977-12	Shift Solenoid B Control Circuit High - Circuit short to battery	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0977-19	Shift Solenoid B Control Circuit High - Circuit current above threshold	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0978-21	Shift Solenoid C Control Circuit Range/Performance - Signal amplitude < minimum	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0978-64	Shift Solenoid C Control Circuit Range/Performance - Signal plausibility failure	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0978-93	Shift Solenoid C Control Circuit Range/Performance - No operation	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0978-94	Shift Solenoid C Control Circuit Range/Performance - Unexpected operation	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0979-11	Shift Solenoid C Control Circuit Low - Circuit short to ground	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0979-14	Shift Solenoid C Control Circuit Low - Circuit short to ground or open	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0980-12	Shift Solenoid C Control Circuit High - Circuit short to battery	■ Transmission control module internal failure	■ Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0980-19	Shift Solenoid C Control Circuit High - Circuit current above threshold	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0981-64	Shift Solenoid D Control Circuit Range/Performance - Signal plausibility failure	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0981-93	Shift Solenoid D Control Circuit Range/Performance - No operation	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0981-94	Shift Solenoid D Control Circuit Range/Performance - Unexpected operation	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0982-11	Shift Solenoid D Control Circuit Low - Circuit short to ground	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0982-14	Shift Solenoid D Control Circuit Low - Circuit short to ground or open	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0982-18	Shift Solenoid D Control Circuit Low - Circuit current below threshold	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0983-12	Shift Solenoid D Control Circuit High - Circuit short to battery	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0983-19	Shift Solenoid D Control Circuit High - Circuit current above threshold	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0984-64	Shift Solenoid E Control Circuit Range/Performance - Signal plausibility failure	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0984-93	Shift Solenoid E Control Circuit Range/Performance - No operation	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0984-94	Shift Solenoid E Control Circuit Range/Performance - Unexpected operation	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0985-11	Shift Solenoid E Control Circuit Low - Circuit short to ground	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0985-14	Shift Solenoid E Control Circuit Low - Circuit short to ground or open	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0985-18	Shift Solenoid E Control Circuit Low - Circuit current below threshold	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0986-12	Shift Solenoid E Control Circuit High - Circuit short to battery	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P0986-19	Shift Solenoid E Control Circuit High - Circuit current above threshold	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P164C-62	Internal Control Module Start-Stop Performance - Signal compare failure	NOTE: This <u>DTC</u> is only active with vehicles installed with stop/start functionality Engine system fault (auto stop/start system)	- Using the Jaguar Land Rover approved diagnostic equipment, perform an ECO-start and monitor the engine speed and engine/transmission start/stop and position signals - Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
P1706-94	High Vehicle Speed Observed in Park - Unexpected operation	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P1707-72	Transfer Case Neutral or Park/Neutral Indication Circuit - Actuator stuck open	- Emergency park release deployed - Parklock mechanism failure - Transmission control module internal failure	- Check the integrity of the emergency park release. Rectify as necessary - Check the integrity of the parklock mechanism. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. Check the parklock mechanism by engaging and disengaging the parklock several times. Rectify as necessary - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P1707-74	Transfer Case Neutral or Park/Neutral Indication Circuit - Actuator slipping	- Emergency park release deployed - Parklock mechanism failure - Transmission control module internal failure	- Check the integrity of the emergency park release. Rectify as necessary - Check the integrity of the parklock mechanism. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. Check the parklock mechanism by engaging and disengaging the parklock several times. Rectify as necessary - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P1707-77	Transfer Case Neutral or Park/Neutral Indication Circuit - Commanded position not reachable	- Emergency park release deployed - Parklock mechanism failure - Transmission control module internal failure	- Check the integrity of the emergency park release. Rectify as necessary - Check the integrity of the parklock mechanism. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. Check the parklock mechanism by engaging and disengaging the parklock several times. Rectify as necessary - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P1707-94	Transfer Case Neutral or Park/Neutral Indication Circuit - Unexpected operation	- Emergency park release deployed - Parklock mechanism failure - Transmission control module internal failure	- Check the integrity of the emergency park release. Rectify as necessary - Check the integrity of the parklock mechanism. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. Check the parklock mechanism by engaging and disengaging the parklock several times. Rectify as necessary - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P177A-07	Transmission Friction Element A or B - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P177B-07	Transmission Friction Element A or C - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P177C-07	Transmission Friction Element A or D - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P177D-07	Transmission Friction Element A or E - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P177E-07	Transmission Friction Element B or C - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P177F-07	Transmission Friction Element B or D - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P178A-07	Transmission Friction Element B or E - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P178B-07	Transmission Friction Element C or D - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P178C-07	Transmission Friction Element C or E - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P178D-07	Transmission Friction Element D or E - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P215B-62	Vehicle Speed / Output Shaft Speed Correlation - Signal compare failure	- Incorrect differential installed - Anti-lock brake system fault - Anti-lock brake system is not configured correctly	- Check the part numbers of the differentials. Install a new differential as necessary - Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index - Using the Jaguar Land Rover approved diagnostic equipment, re-configure the anti-lock brake system control module with the latest level software
P258F-02	Torque Management Request Output Signal - General signal failure	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P2638-67	Torque Management Feedback Signal A Range/Performance - Signal incorrect after event	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P2700-07	Transmission Friction Element A Apply Time Range/Performance - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P2701-07	Transmission Friction Element B Apply Time Range/Performance - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2702-07	Transmission Friction Element C Apply Time Range/Performance - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P2703-07	Transmission Friction Element D Apply Time Range/Performance - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P2704-07	Transmission Friction Element E Apply Time Range/Performance - Mechanical failures	Automatic transmission internal failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, install a new automatic transmission
P2711-94	Unexpected Mechanical Gear Disengagement - Unexpected operation	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P2713-04	Pressure Control Solenoid D - System internal failures	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P2722-04	Pressure Control Solenoid E - System internal failures	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P2757-93	Torque Converter Clutch Pressure Control Solenoid Control Circuit Perf or Stuck Off - No operation	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P2757-94	Torque Converter Clutch Pressure Control Solenoid Control Circuit Perf or Stuck Off - Unexpected operation	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P2759-11	Torque Converter Clutch Pressure Control Solenoid Electrical - Circuit short to ground	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P2759-12	Torque Converter Clutch Pressure Control Solenoid Electrical - Circuit short to battery	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P2761-13	Torque Converter Clutch Pressure Control Solenoid Control Circuit/Open - Circuit open	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P2762-64	Torque Converter Clutch Pressure Control Solenoid Control Circuit Range/Perf - Signal plausibility failure	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2784-02	Input Shaft Speed Sensor A/B Correlation - General signal failure	- Engine system fault - Anti-lock brake system fault	- Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index - Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
P2787-4B	Clutch Temperature Too High - Over temperature	Automatic transmission temperature above maximum threshold	NOTE: This <u>DTC</u> may be induced by performing an excessive number of gear shifts using the paddle switches. Allow the system temperature to return to normal. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Check the paddle switches for uncommanded operation. Rectify as necessary
P2787-98	Clutch Temperature Too High - Component or system over temperature	Automatic transmission temperature above maximum threshold	NOTE: This <u>DTC</u> may be induced by performing an excessive number of gear shifts using the paddle switches. Allow the system temperature to return to normal. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and perform a road test. If the fault persists, refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Check the paddle switches for uncommanded operation. Rectify as necessary. Clear the <u>DTCs</u> and perform a road test. If the fault persists, or if any other gear ratio related <u>DTCs</u> are set, install a new automatic transmission
P2793-92	Gear Shift Direction Circuit - Performance or incorrect operation	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
P2793-94	Gear Shift Direction Circuit - Unexpected operation	Transmission control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module
U0001-81	High Speed <u>CAN</u> Communication Bus - Invalid serial data received	Invalid data received from another control module on the high speed <u>CAN</u> bus (powertrain)	Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0001-82	High Speed <u>CAN</u> Communication Bus - Alive/sequence counter incorrect / not updated	Invalid data received from another control module on the high speed <u>CAN</u> bus (powertrain)	Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0001-83	High Speed <u>CAN</u> Communication Bus - Value of signal protection calculation incorrect	Invalid data received from another control module on the high speed <u>CAN</u> bus (powertrain)	Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0001-86	High Speed <u>CAN</u> Communication Bus - Signal invalid	Invalid signal received from another control module on the high speed <u>CAN</u> bus (powertrain)	Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid signal source control module. Check the relevant control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0001-87	High Speed <u>CAN</u> Communication Bus - Missing message	Missing message from another control module on the high speed <u>CAN</u> bus (powertrain)	Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the missing message source control module. Check the relevant control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0073-00	Control Module Communication Bus A Off - No sub type information	High speed <u>CAN</u> bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the Jaguar Land Rover approved diagnostic equipment, perform a <u>CAN</u> network integrity test. Refer to the electrical circuit diagrams and check the high speed <u>CAN</u> bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
U0121-87	Lost Communication With Anti-Lock Brake System (ABS) Control Module - Missing message	Missing/invalid data from the anti-lock brake system control module	Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0300-57	Internal Control Module Software Incompatibility - Invalid/incomplete software component	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the <u>DTCs</u>. Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the <u>DTCs</u> and retest
U0401-81	Invalid Data Received From ECM/PCM A - Invalid serial data received	Missing/invalid data from the powertrain control module	Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0401-82	Invalid Data Received From ECM/PCM A - Alive/sequence counter incorrect / not updated	Missing/invalid data from the powertrain control module	Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0401-83	Invalid Data Received From ECM/PCM A - Value of signal protection calculation incorrect	Missing/invalid data from the powertrain control module	Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0401-87	Invalid Data Received From ECM/PCM A - Missing message	Missing/invalid data from the powertrain control module	Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0415-82	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Alive/sequence counter incorrect / not updated	Missing/invalid data from the anti-lock brake system control module	Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0415-83	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Value of signal protection calculation incorrect	Missing/invalid data from the anti-lock brake system control module	Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U2101-56	Control Module Configuration Incompatible - Invalid/incomplete configuration	- Car configuration file mismatch with vehicle specification - Transmission control module is not configured correctly	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the <u>DTCs</u>. - Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the <u>DTCs</u> and retest - Using the Jaguar Land Rover approved diagnostic equipment, re-configure the transmission control module with the latest level software
U3000-56	Control Module - Invalid/incomplete configuration	- Transmission control module is not configured correctly - Transmission control module internal failure	- Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the transmission control module with the latest level software - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transmission control module. Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Learn <u>VIN</u> (0x0404) - - If the fault persists, refer to ZF Field Service
U3000-9A	Control Module - Component or system operating conditions	Engine system fault	Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U3001-94	Control Module Improper Shutdown - Unexpected operation	- Transmission control module power or ground circuit open circuit, high resistance - Battery/charging system fault	- Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary - Refer to the relevant section of the workshop manual and check the battery and charging system. Rectify as necessary